

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Polynomials

Use factors, remainders, and roots as the language of polynomial questions.

POLYNOMIALS · P1 · 05

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 REMA: Remainder Theorem

TRIGGER *remainder when divided by $x - a$*

02 FACTOR: Factor Theorem

TRIGGER *show $x - a$ is a factor*

03 DIVIDE: Polynomial Division

TRIGGER *divide by a linear or quadratic factor*

04 ROOTS: Use Known Roots

TRIGGER *roots, factorise completely*

BANK EVIDENCE Local pages: 11 problem, 21 answer, 6 notes. Website primary entries: 4.

01 REMA: Remainder Theorem

TRIGGER *remainder when divided by $x - a$*

BECOMES Substitute $x = a$.

FIRST LINE TO WRITE

$$\text{remainder} = f(a)$$

SIMPLEST STRATEGY

- 1 Read the divisor $x - a$.
- 2 Evaluate $f(a)$.
- 3 Match this value to the remainder.
- 4 Answer without full division.

WORKED MODEL

| If divisor is $x - 2$, remainder is $f(2)$.

HARD VARIANTS

- 1 For divisor $ax - b$, substitute $x = b/a$, not just b .
- 2 If the remainder is given as a condition, set $f(b/a)$ equal to that value.
- 3 Use substitution when long division would waste time.

— BOTTOM LINE

For $x - a$, the remainder is $f(a)$.

02 FACTOR: Factor Theorem

TRIGGER *show $x - a$ is a factor*

BECOMES A zero remainder proves a factor.

FIRST LINE TO WRITE

$$f(a) = 0$$

SIMPLEST STRATEGY

- 1 Find the value from the divisor.
- 2 Apply substitution.
- 3 Check for zero.
- 4 Tell the conclusion.
- 5 Obtain the remaining factor.
- 6 Roots follow from factors.

WORKED MODEL

$$f(3) = 0 \Rightarrow (x - 3) \text{ is a factor.}$$

HARD VARIANTS

- 1 For $ax - b$, prove $f(b/a) = 0$ before claiming the factor.
- 2 If a parameter appears, use the zero-remainder condition to find it.
- 3 After proving one factor, divide to reduce the polynomial.

— BOTTOM LINE

A factor is just a divisor with zero remainder.

03 DIVIDE: Polynomial Division

TRIGGER *divide by a linear or quadratic factor*

BECOMES Long division or comparison of coefficients.

FIRST LINE TO WRITE

$$f(x) = (\text{divisor})(\text{quotient}) + \text{remainder}$$

SIMPLEST STRATEGY

- 1 Decide quotient degree.
- 2 Insert unknown coefficients.
- 3 Value-match by expanding.
- 4 Identify the quotient.
- 5 Declare remainder.
- 6 Extract roots if possible.

WORKED MODEL

$$x^3 + 2x^2 - x - 2 = (x + 2)(x^2 - 1).$$

HARD VARIANTS

- 1 If the quotient has unknown coefficients, compare powers of x after expanding.
- 2 Keep the remainder term until the final comparison is complete.
- 3 Use the reduced polynomial to find remaining roots.

— BOTTOM LINE

Comparison of coefficients often beats long division.

04 ROOTS: Use Known Roots

TRIGGER *roots, factorise completely*

BECOMES Each root gives a factor.

FIRST LINE TO WRITE

$$x = a \Rightarrow (x - a)$$

SIMPLEST STRATEGY

- 1 Record each root.
- 2 Obtain its factor.
- 3 Organise the product.
- 4 Test the leading coefficient.
- 5 State all roots/factors.

WORKED MODEL

| Roots 2, -1 give factors $(x - 2)(x + 1)$.

HARD VARIANTS

- 1 Repeated roots create repeated factors; do not list them once if multiplicity matters.
- 2 If roots are surds or complex-looking pairs, use conjugate/symmetric factors.
- 3 Match the leading coefficient before expanding the final polynomial.

— BOTTOM LINE

Roots and factors are the same information in different forms.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
remainder when divided by $x - a$	$\text{remainder} = f(a)$
show $x - a$ is a factor	$f(a) = 0$
divide by a linear or quadratic factor	$f(x) = (\text{divisor})(\text{quotient}) + \text{remainder}$
roots, factorise completely	$x = a \Rightarrow (x - a)$